

L2

GC Code

e2h2zotc

Human Motor control

- **Computational:** Mathematical calculations to achieve a movement
- **Procedural:** The behavioural and cognitive states to execute a movement.
- **Implementation:** How muscles contract/functions. The part of brain contributing in execution of a movement.

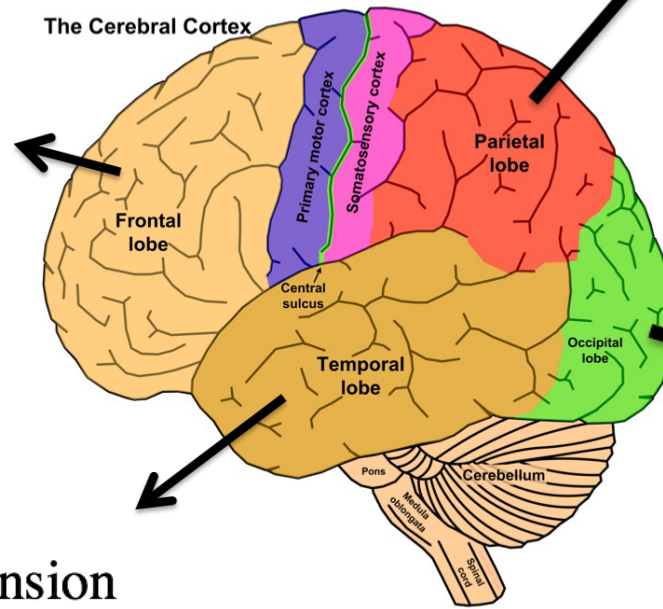
Interdisciplinary

David Marr Tri-level hypothesis

Cerebral Cortex

- Decision making
- Reward
- Speech and Language Control
- Memory formation

- Audition
- Face recognition
- Speech comprehension
- Color perception



- Attention
- Eye movement
- Heading direction
- Spatial location

- Vision
- Depth perception

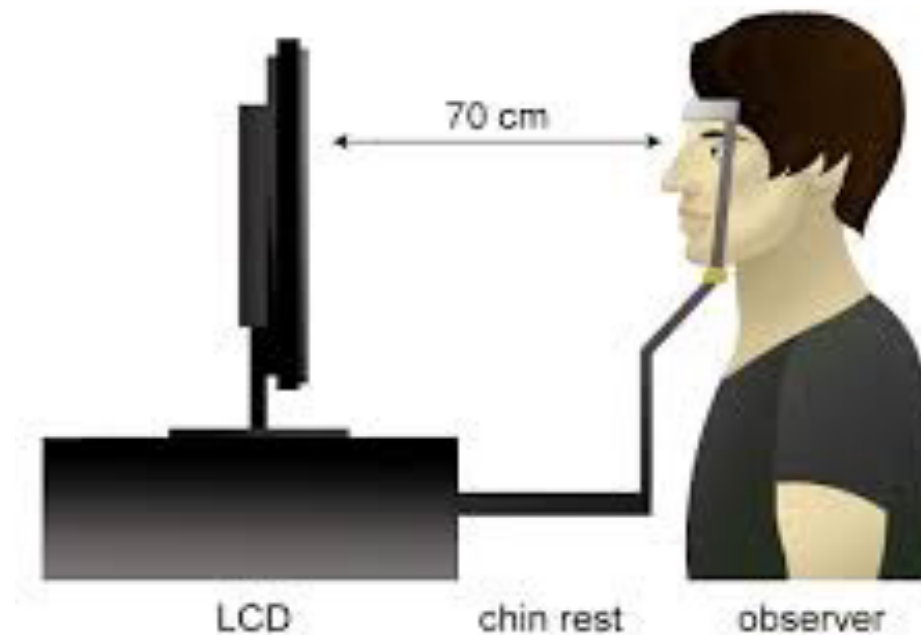
Psychology

Study of Mind → Study of Behaviour → Psychophysics

Technologies



Cognitive Science: Interdisciplinary study of mind and its processes.



Brain

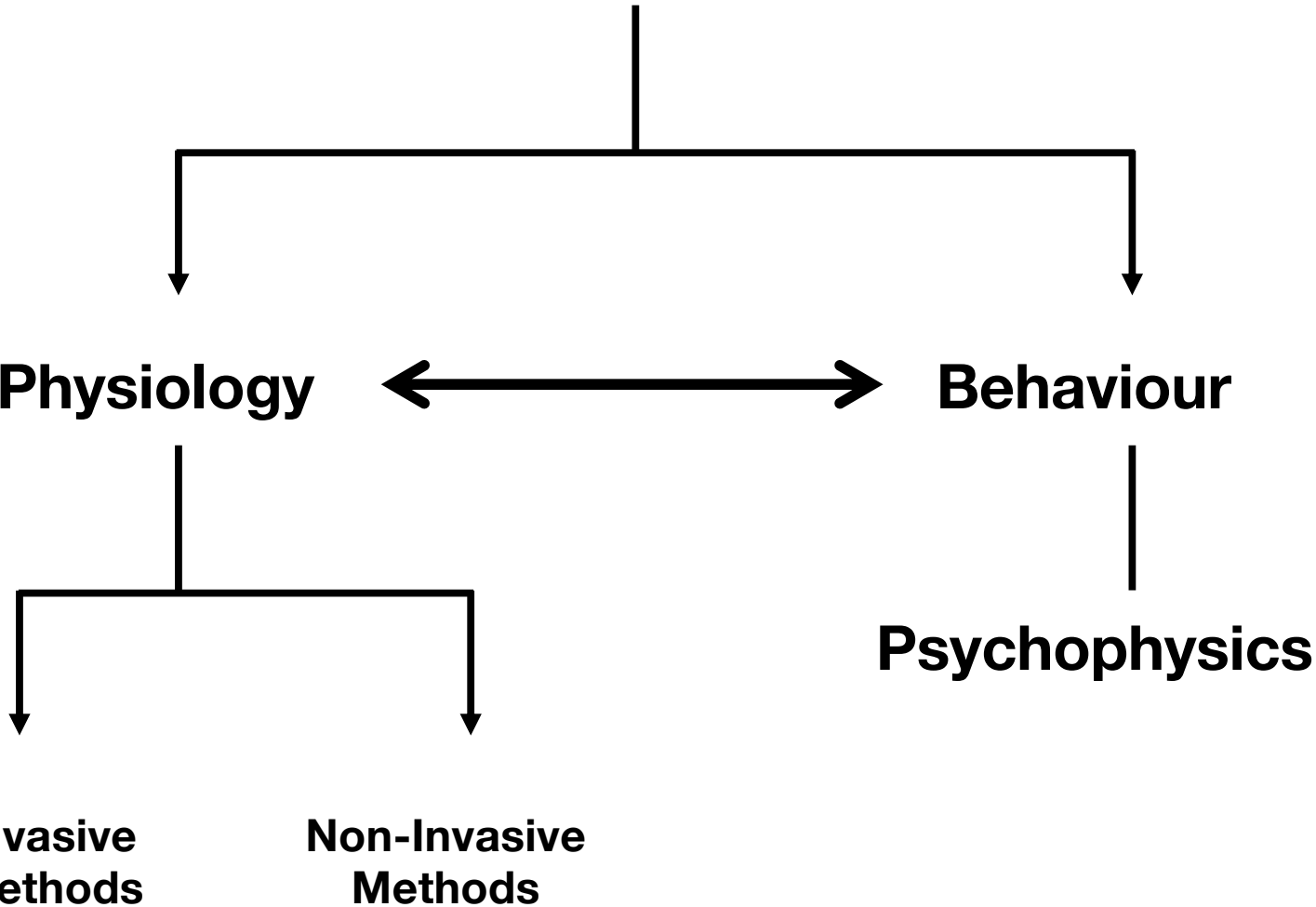
Physiology

Behaviour

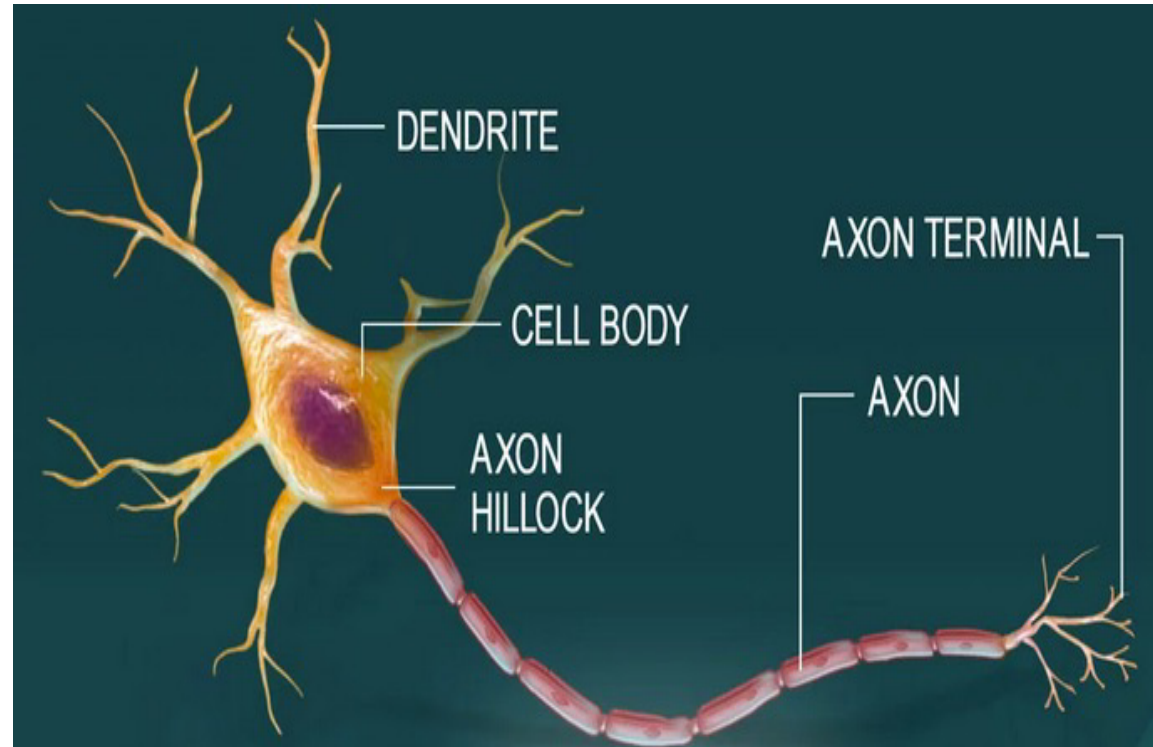
Psychophysics

**Invasive
Methods**

**Non-Invasive
Methods**

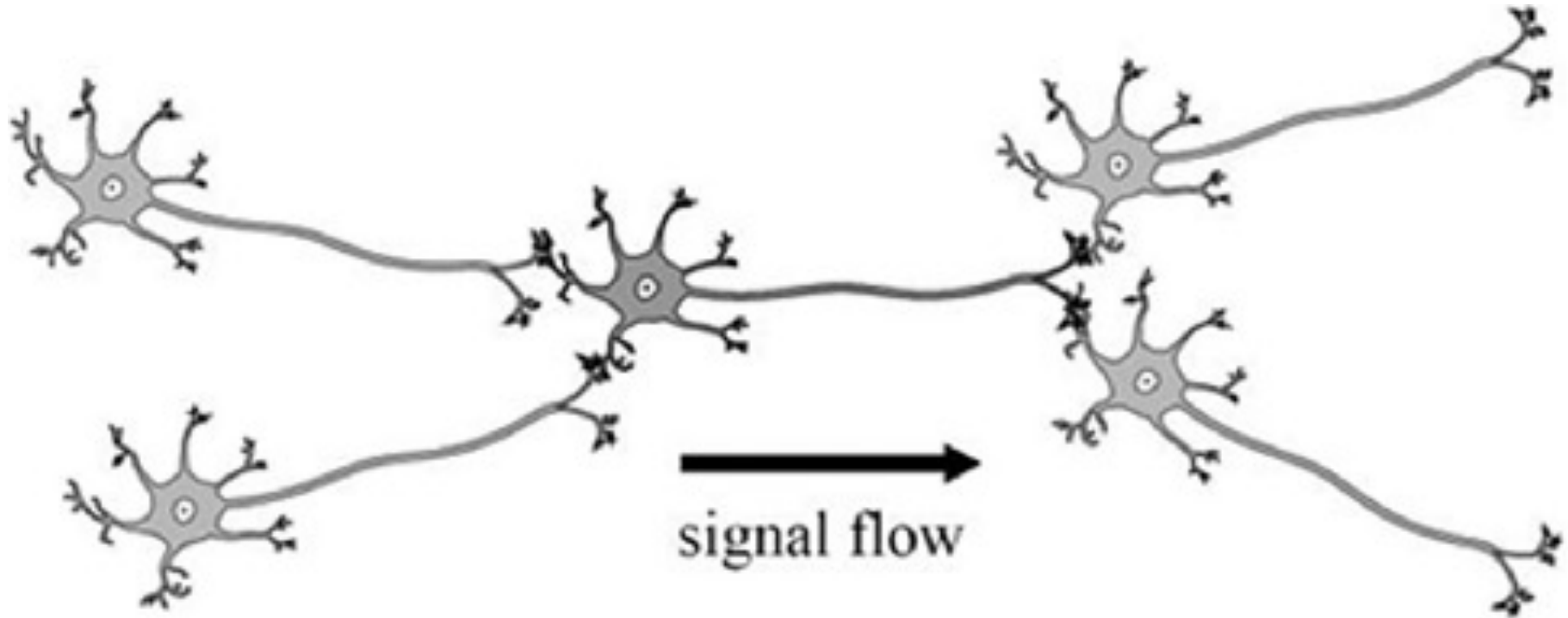


Neurons



<https://www.visiblebody.com/learn/nervous/neurons>

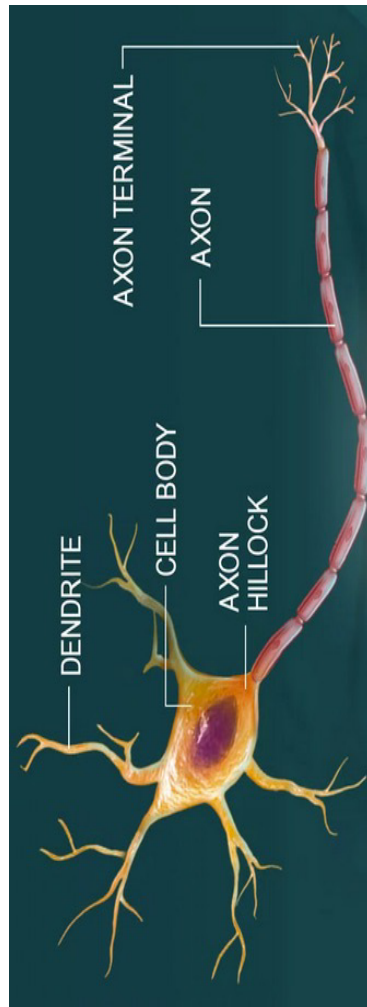
Neural Networks



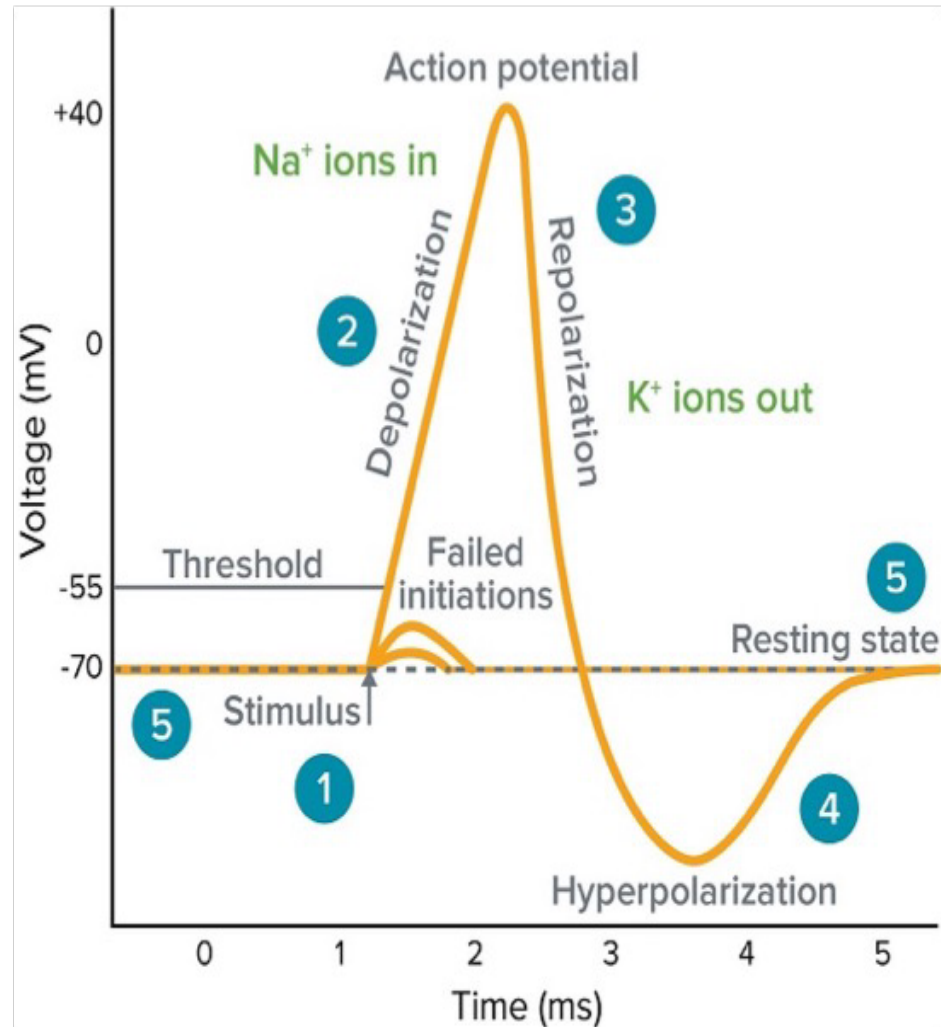
How can we study neural activities?

Action Potential

Action Potential: A transient change in the potential of the cell membrane when it is exposed to a stimulus



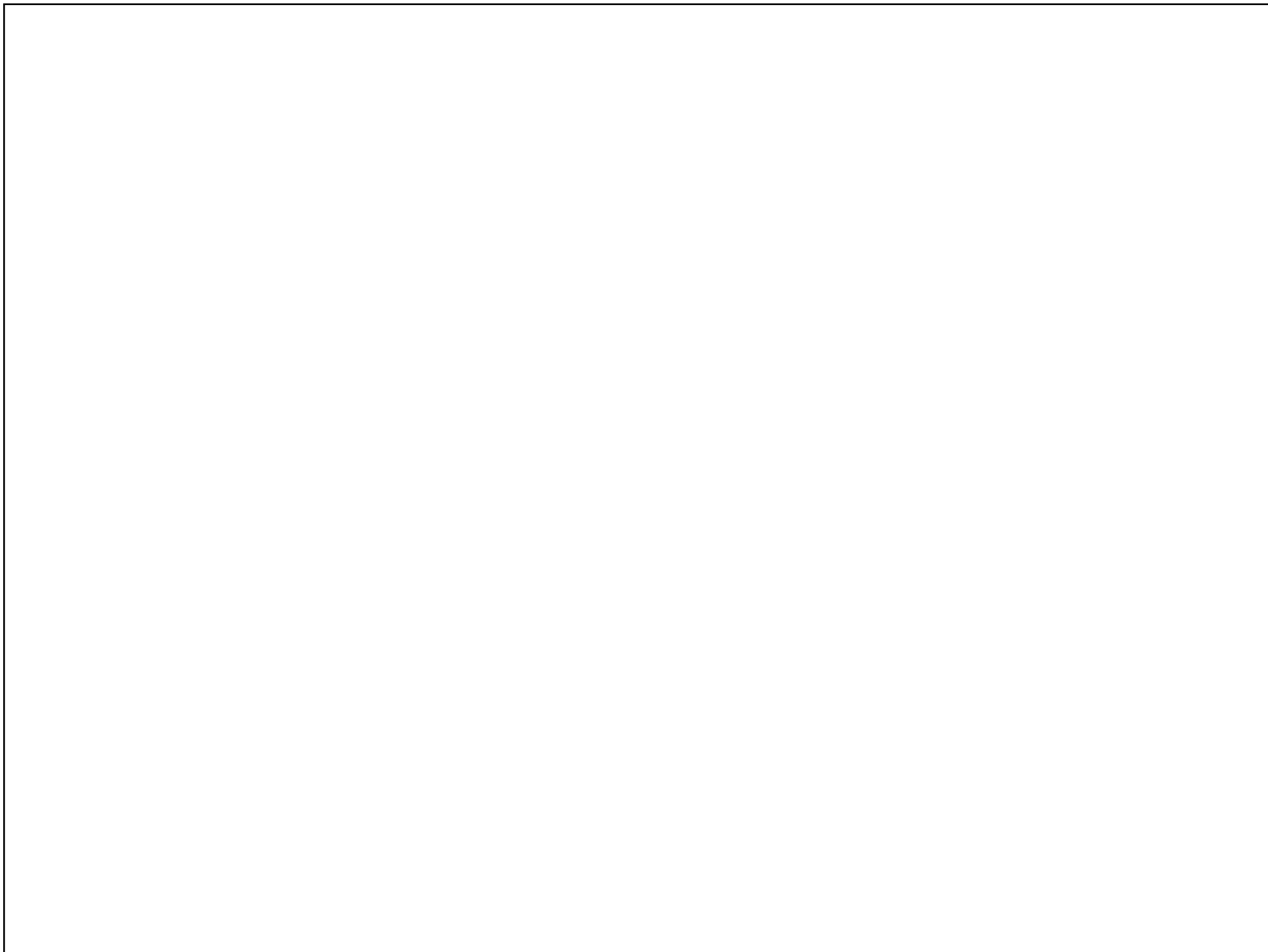
<https://www.visiblebody.com/learn/nervous/neurons>



<https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential#gref>

Activity

Report presence of a circular stimulus at the
center of the screen





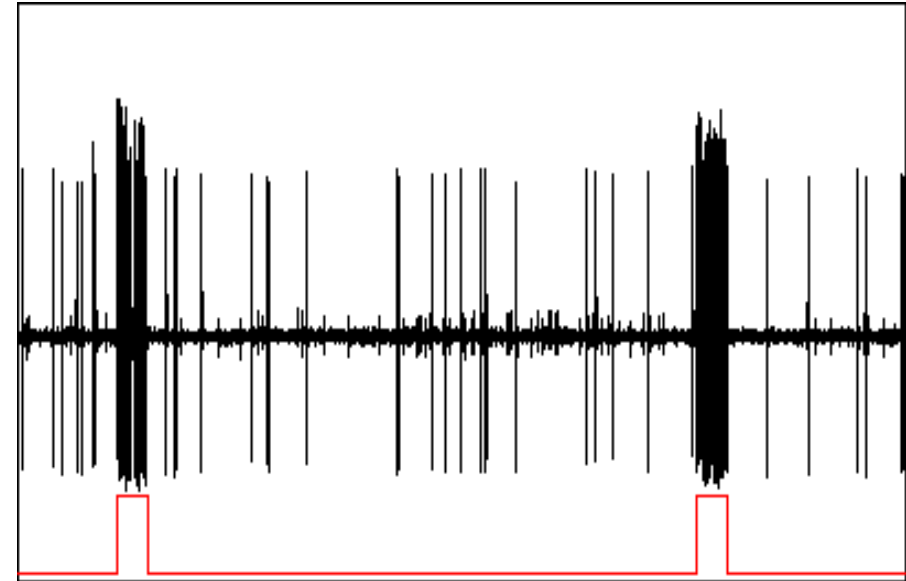
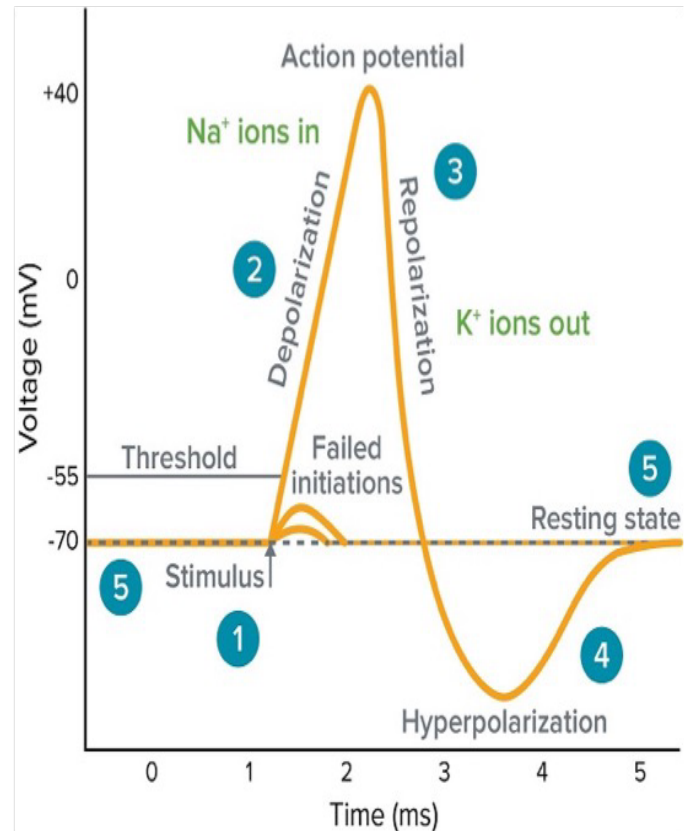








Action Potential



<https://www.ece.rice.edu/~dhj/spike.html>

<https://www.moleculardevices.com/applications/patch-clamp-electrophysiology/what-action-potential#ref>

Language of brain

How can we measure Action potentials?

Methodologies

Brain

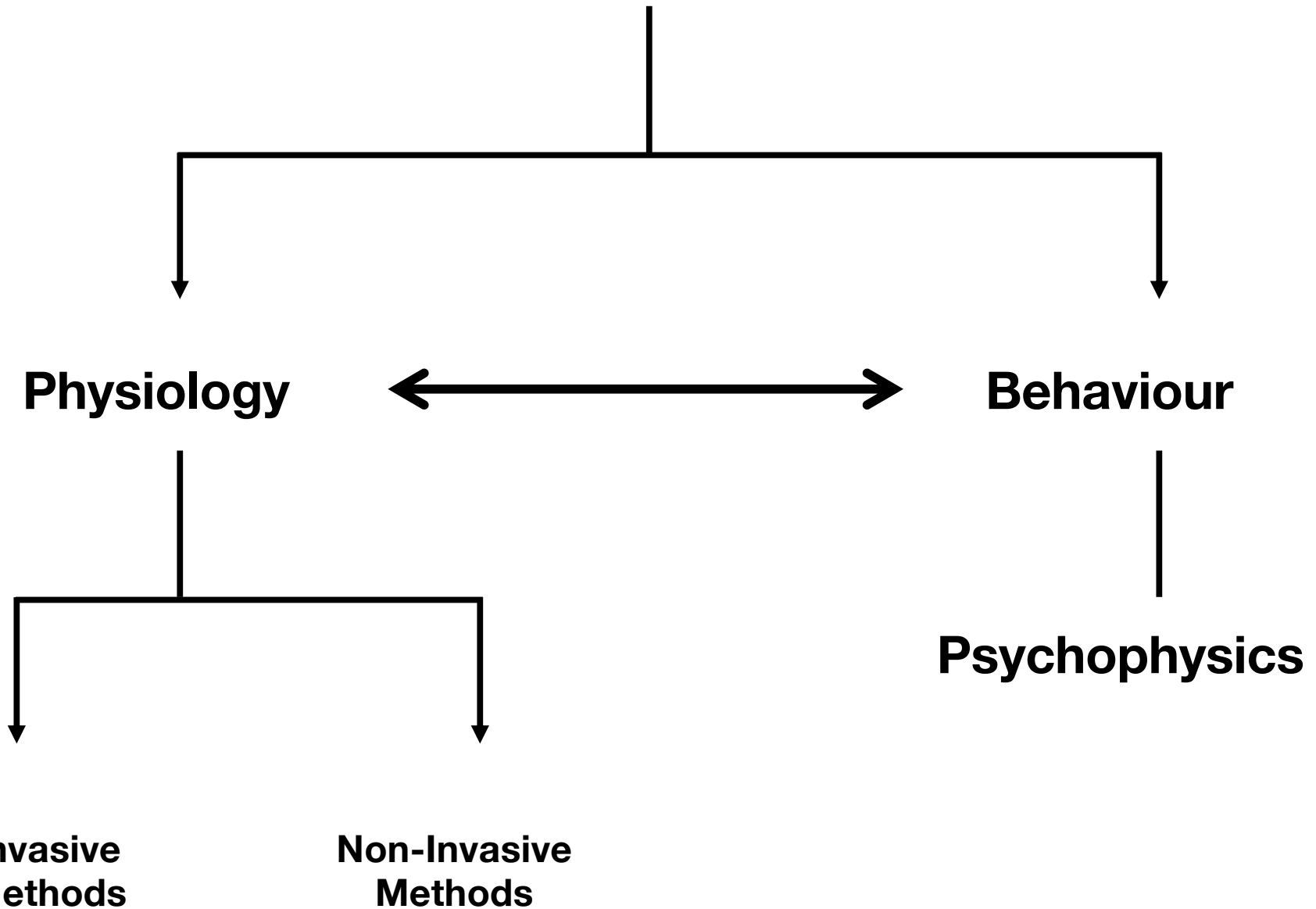
Physiology

Behaviour

Psychophysics

**Invasive
Methods**

**Non-Invasive
Methods**



The 3 R's of Animal Research

Reduce



Reduce the number of animals used

Refine



Refine tests to cause animals less stress

Replace

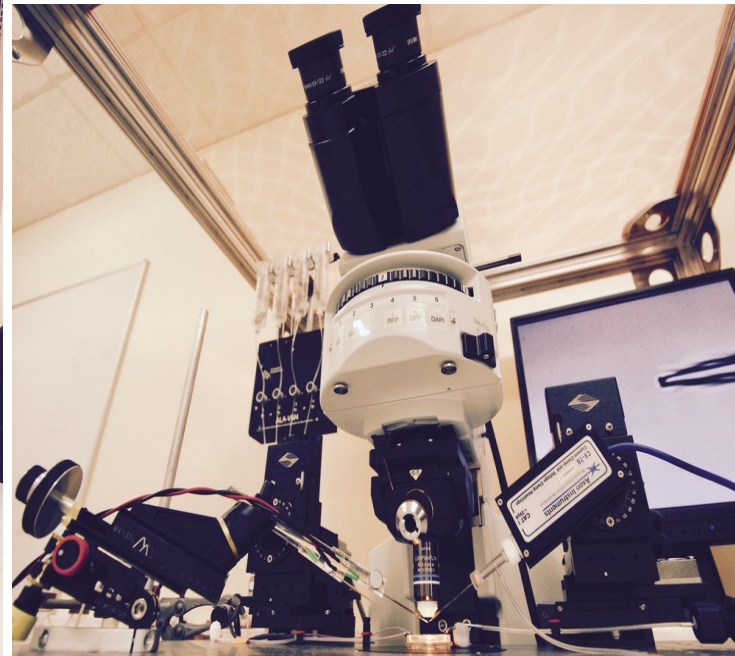
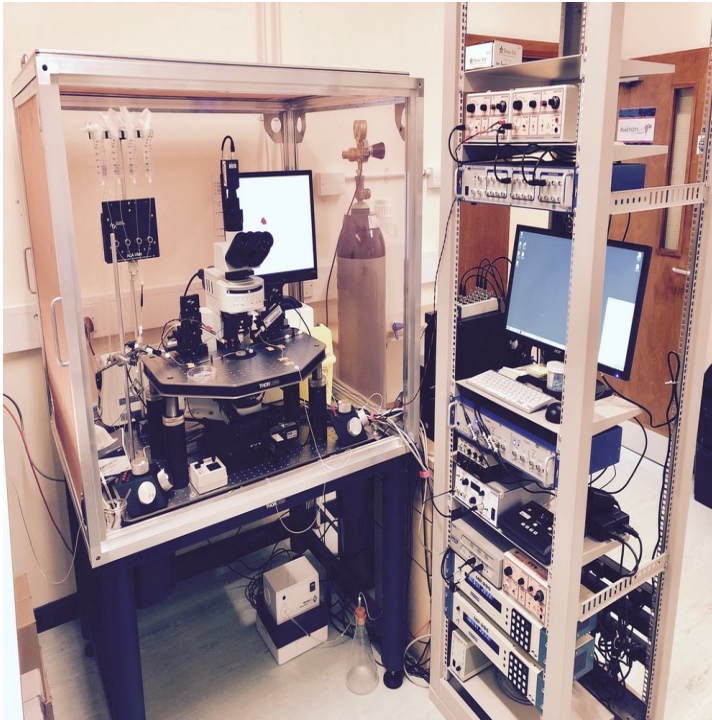


Replace animal studies with other methods

Invasive Methodologies

Intracellular recording

Recording action potential from an individual neuron.



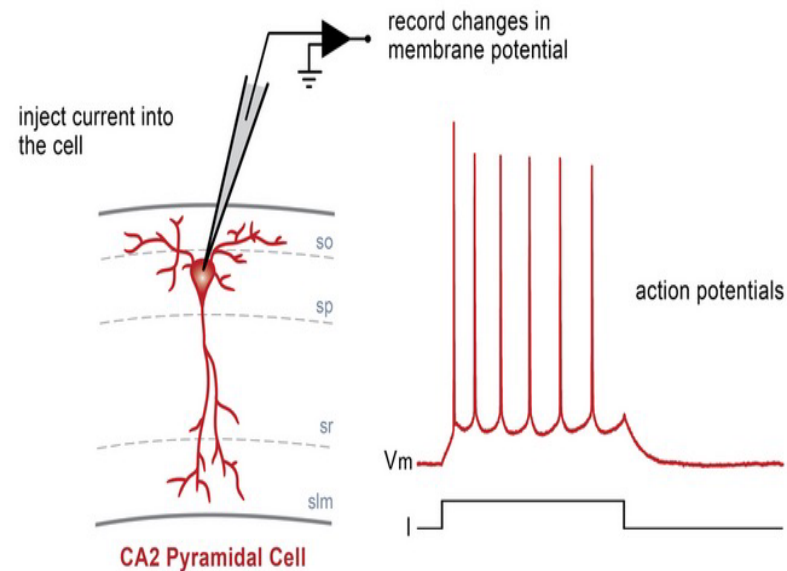
<https://www.plasticitylab.com/methods>

Intracellular recording

Studying membrane dynamics



<https://dagemti.com/dage-mti-ir-1000-real-time-infrared-microscopy-camera/>

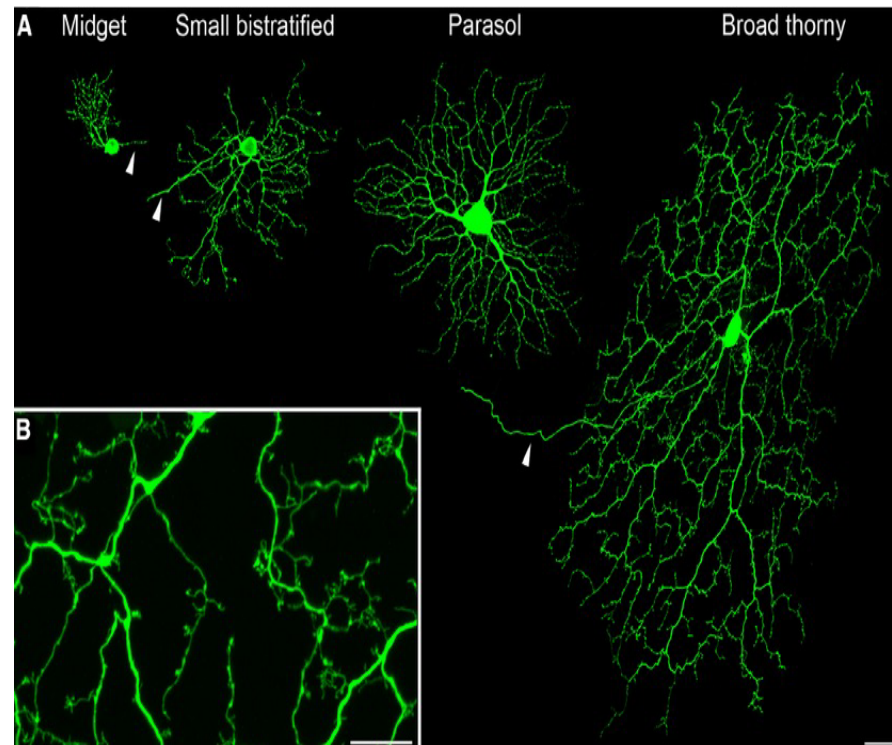


<https://www.plasticitylab.com/methods>

Measuring voltage or current across a membrane of a cell.

Intracellular recording

Studying neuronal morphology

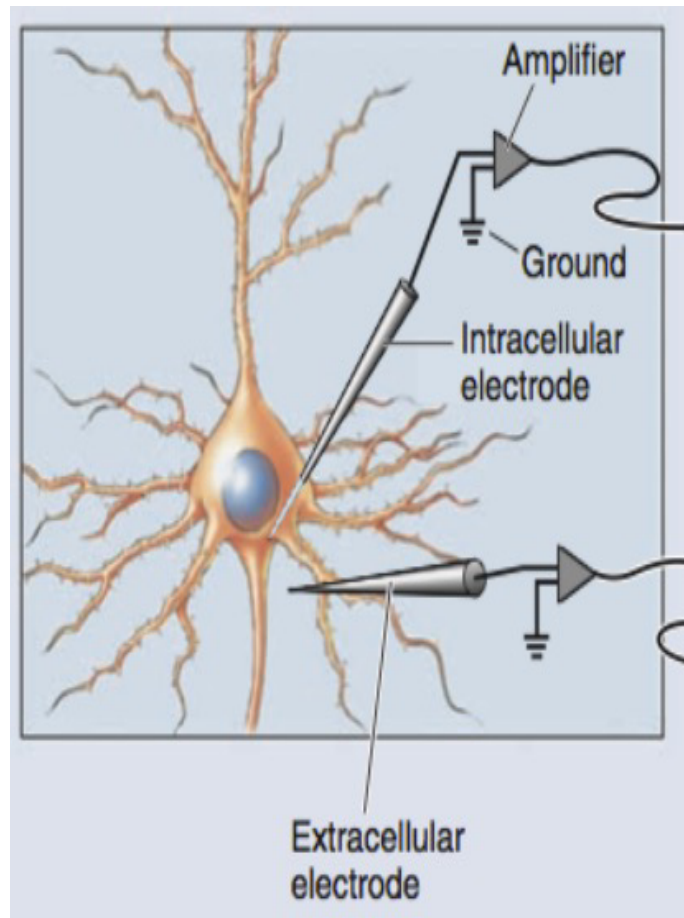


Puller et al, 2015

Restricted to one neuron at a time.

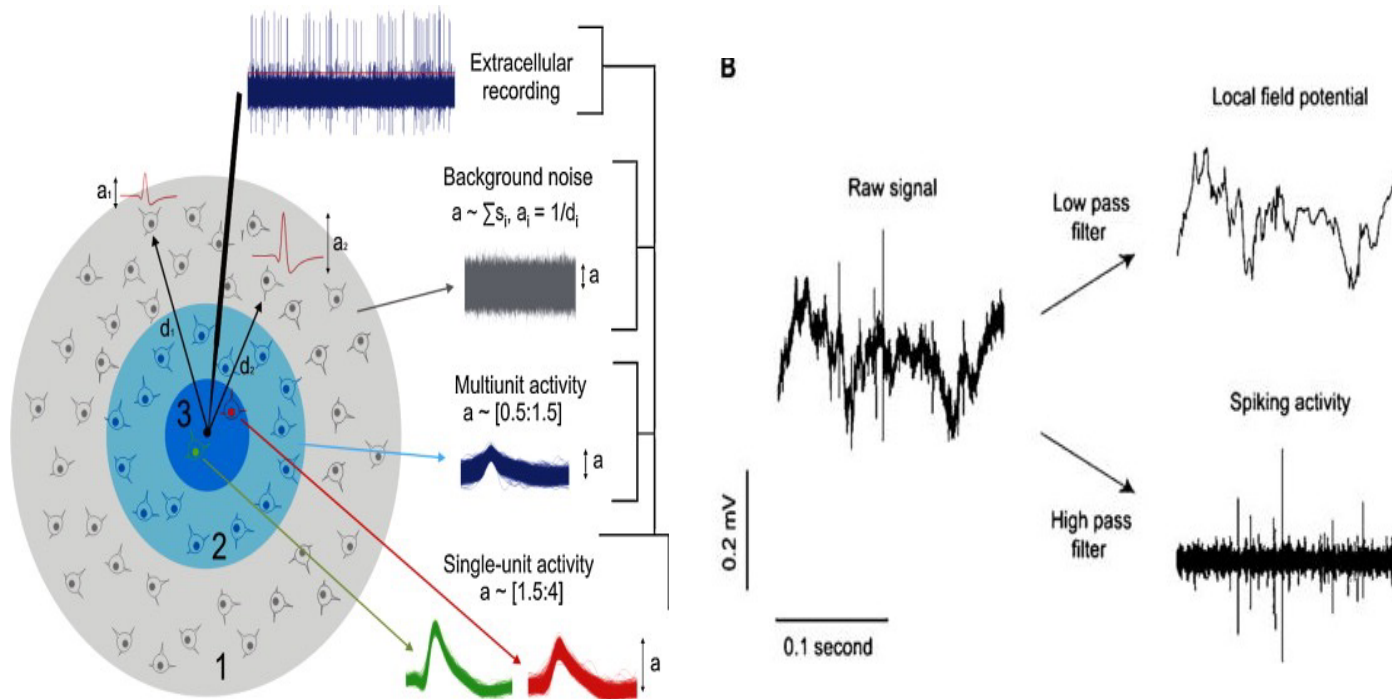
Extracellular recording

Neuronal activity measured in area adjacent to a neuron.



- Single Unit
- Multi-Unit
- Local Field Potential

Extracellular recording



Bozer et al, 2017

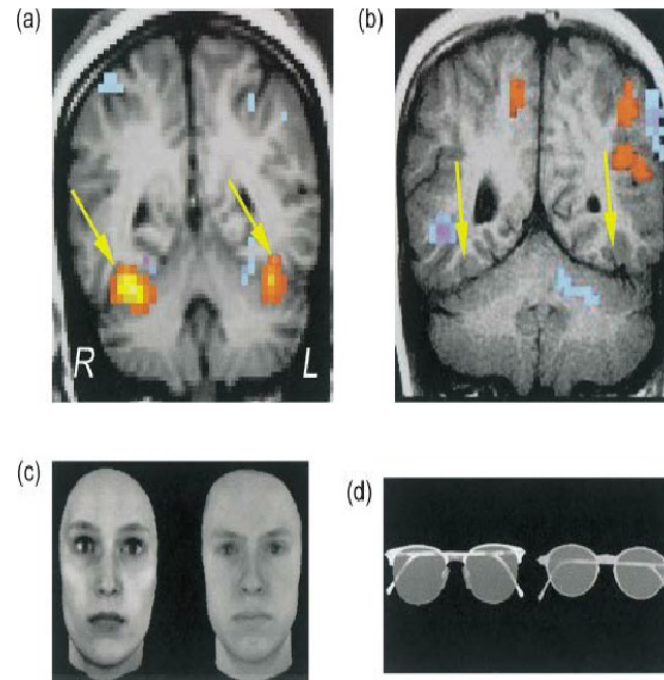
<http://charlesfrye.github.io/FoundationalNeuroscience/80/>

Local field potentials (LFP) are transient electrical signals generated in nervous and other tissues by the summed and synchronous electrical activity of the individual cells (e.g. neurons) in that tissue (*Wikipedia*).

Restricted to small number of neurons in an area of brain.

fMRI(functional Magnetic Resonance Imaging)

<https://www.youtube.com/watch?v=nvB9hAarzw4>



Grelotti et al, 2001

Measures brain activity associated with blood flow.

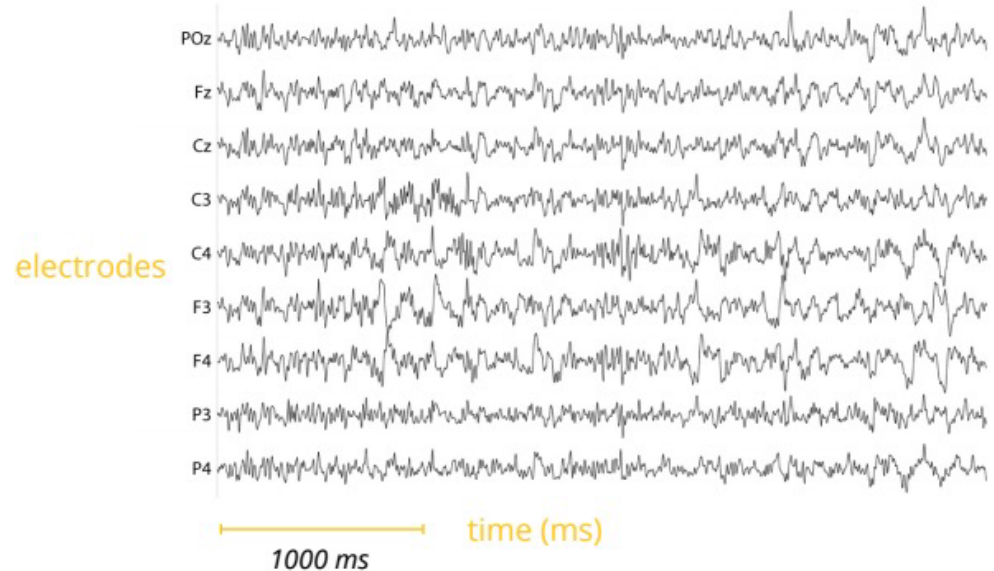
fMRI(functional Magnetic Resonance Imaging)

- **Measures blood oxgenation levels**
 - **Good spatial resolution**
 - **Bad temporal resolution**

EEG (Electroencephalography)



<https://nld.tamu.edu/eeeg>



<https://imotions.com/blog/eeeg/>

Non-invasive electrophysiological method to record brain activity.

EEG reflects summed post-synaptic activities of large number of cells.

EEG (Electroencephalography)

- **Measures voltage fluctuations resulting from ionic current within the neurons of the brain.**
- **Bad spatial resolution**
- **Good temporal resolution**